Week 4 Problem Set: The Squeeze Theorem and Indeterminate Forms

Calculus Honors · Module 2: Limits Without Bound & Special Limits · Lessons 2.03 and 2.04

Name: _____ Date: _____

> Show all of your work. A correct answer with no reasoning earns partial credit at most. Keep your writing neat and vertical, line up your equal signs, and label your final answers.

PART A · THE SQUEEZE THEOREM

1. Suppose $-x^2 \le f(x) \le x^2$ for all x near 0. Find $\lim_{x \to 0} f(x)$ using the Squeeze Theorem, and name which hypothesis of the theorem makes the conclusion valid.

2. Suppose $2 - x^2 \le g(x) \le 2 + x^2$ for all x near 0. Evaluate $\lim_{x \to 0} g(x)$.

3. Show that $\lim_{x \to 0} x^2 \cos(1/x) = 0$ using the Squeeze Theorem. State the bounding inequality you use, starting from the fact that $-1 \le \cos(1/x) \le 1$ for every $x \neq 0$.

PART B · THE SPECIAL LIMIT AND ITS RELATIVES

4. Use $\lim_{x\to 0}\dfrac{\sin x}{x}=1$ to evaluate each. (a) $\lim_{x\to 0}\dfrac{\sin 3x}{x}$ (b) $\lim_{x\to 0}\dfrac{\sin 2x}{\sin 5x}$ (c) $\lim_{x\to 0}\dfrac{x}{\sin x}$

PART C · CLASSIFY AND RESOLVE INDETERMINATE FORMS

5. For each limit, state which indeterminate form it presents (if any), and evaluate it if algebra or a special limit you know resolves it. If it does not resolve with tools you have, say so and name the general tool that will eventually apply. (a) $\lim_{x\to 4}\dfrac{x^2-16}{x-4}$ (b) $\lim_{x\to\infty}\dfrac{6x^2+1}{3x^2-x}$ (c) $\lim_{x\to 0}\dfrac{\sqrt{x+16}-4}{x}$ (d) $\lim_{x\to 0}\dfrac{x-\sin x}{x^3}$

PART D · EXPLAIN YOUR THINKING

6. Explain in your own words why "indeterminate" does not mean "unsolvable." Use one worked example from this module as evidence.